Business Management HL · Set B · Paper 2

Question booklet · 105 minutes · 50 marks

Answer every compulsory question and choose one complete Section B option. A calculator is permitted. Use the case evidence and show calculations. Write on separate paper.

Section A · Bracken Sensors [12]

Bracken assembles monitoring units. Normal annual capacity is 50,000 units and output is 40,000. Total annual fixed manufacturing overhead is CU 360,000. Each unit sells for CU 48 and uses CU 28 in variable costs. A one-off foreign order for 8,000 units offers CU 35 per unit. Shipping the order would cost CU 16,000 in total. Normal sales can be maintained.

(a) Calculate current capacity utilization and fixed manufacturing overhead per unit if fixed overhead is allocated over actual output. **[4]**

(b) Calculate the change in operating profit if Bracken accepts the order. **[3]**

(c) Explain why an absorption-costing appraisal and a contribution appraisal could lead to different decisions. **[3]**

(d) Explain one reason spare capacity measured annually might overstate Bracken's ability to fulfil this order. **[2]**

Section A · Mora Studio [10]

Mora has non-current liabilities of CU 240,000 and equity of CU 360,000, with no other long-term capital. It earns annual profit before interest and tax of CU 84,000. A project needs CU 150,000 and is forecast to add CU 15,000 to annual profit before interest and tax. It can be financed by new long-term debt or ordinary shares. Annual interest on the new debt would be 8%.

(a) Calculate current gearing and return on capital employed. Use non-current liabilities as a percentage of capital employed for gearing. **[4]**

(b) Calculate gearing after expansion under each financing option. **[3]**

(c) Calculate additional annual profit before tax after interest if Mora borrows. Explain one implication for risk. **[3]**

Section A · Willow Joinery [8]

Willow is planning a small factory refit. Times are in working days. Activity A takes 4 days and B takes 6 days. Both can begin immediately. C takes 5 days after A. D takes 2 days after B. E takes 3 days and can start only after both C and D finish. All durations are deterministic for the initial plan.

(a) Identify the critical path and calculate the minimum project duration. **[3]**

(b) Determine total float on B and the effect of a two-day delay to B on overall completion. **[3]**

(c) Explain one reason the critical path could change during the refit. **[2]**

Section B · Option 1 · Senna Lamps [20]

Senna makes 12,000 lamps annually at CU 45 each. Operating profit is CU 54,000. Inventory averages CU 36,000, cost of sales CU 288,000 and trade payables CU 24,000. All CU 240,000 of annual purchases are on credit. Assume a 360-day year. A new supplier would deliver smaller batches, reducing average inventory to CU 24,000. Delivery charges of CU 4,800 a year would be new operating expenses. Units and cost of sales otherwise stay constant. The supplier has a single local depot.

(a) Calculate current inventory turnover in days. **[2]**

(b) Calculate current creditor days and inventory days under the proposal. **[4]**

(c) Find the change in the operating profit margin and explain the difference between the cash released from inventory and this change in profit. **[4]**

(d) Evaluate whether Senna should adopt the new supply arrangement. **[10]**

Section B · Option 2 · Fen Rail Events [20]

Fen chooses between a permanent venue and a portable stage. Both options have four-year lives and no scrap value. Net annual cash inflows, constant within each scenario, are shown below. Use a four-year annuity factor of 3.31. Each scenario applies for the full four years. The owner has limited access to borrowing. An outdoor portable stage can be hired at several sites.

Option	Initial cost / CU	Annual inflow if busy / CU	Annual inflow if quiet / CU
Venue	180,000	80,000	35,000
Stage	100,000	45,000	32,000

The estimated probability of the busy scenario is 0.60.

(a) Calculate the NPV of the venue in the busy scenario. **[2]**

(b) Calculate the probability-weighted expected NPV for each option. **[4]**

(c) Calculate both quiet-scenario NPVs and explain how these outcomes inform a risk-averse owner. **[4]**

(d) Recommend an investment, taking account of both risk and the reliability of the figures. **[10]**

End of question booklet · 50 marks